



# RATPAC2 Summer School

Welcoming remarks and brief summer school overview

ratpac-two summer school tutorials

Marc Bergevin

Lawrence Livermore National Laboratory

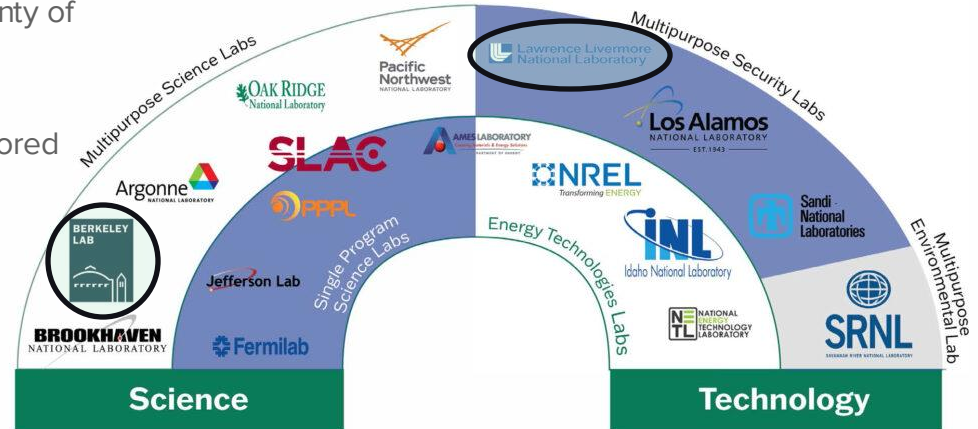
# Welcome to UCLCC!



- What are the expectations for the summer school?
  - By end of the school, to have expertise in how to use RATPAC2 for your needs.
  - The summer school is devised to have plenty of time for hands-on use of this software
  - We have a wide level of expertise among participants; the working exercises are tailored with this in mind.



Livermore  
Collaboration  
Center



\*Ratpac-Two v1.0.0, M. Askins, T. Kaptanoglu, M. Smiley, N. Rowe, M. Bergevin, 2023, <https://doi.org/10.11578/dc.20240620.8>

# RAT, RATPAC, RATPAC2

## *The need for a new Philosophy*



- Early development in the last decade were centered around reactor antineutrino detector technology (BRAIDWOOD, WATCHMAN, now defunct)
  - The light emission model used in software (GLG4Sim) makes it attractive for a wide set of detector technology (Cherenkov, scintillation, optical fiber,...)
  - Software has recently been adapted to scintillation crystal detectors
  - A potential drawback is that it is very very easy to branch off

### Experiments/Testbeds that have used RATPAC

- SNO+
- MiniCLEAN
- DEAP
- JSNS
- nuDOT
- Eos
- Theia
- ANNIE
- BUTTON
- LiquidO/CLOUD
- WATCHBOY
- WATCHMAN/AIT-NEO and LLNL R&D testbeds
- XLZD (under consideration)

### Reactor Analysis Tools (RAT) is born

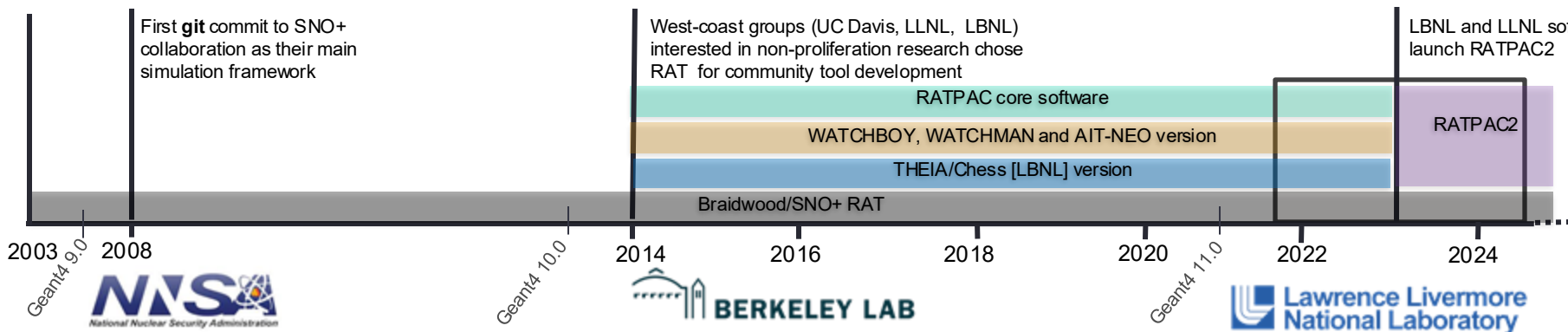
First **git** commit to SNO+ collaboration as their main simulation framework

### Reactor Analysis Tools Plus Additional Code (RATPAC) is born

West-coast groups (UC Davis, LLNL, LBNL) interested in non-proliferation research chose RAT for community tool development

### RATPAC2 is born

LBNL and LLNL soft launch RATPAC2

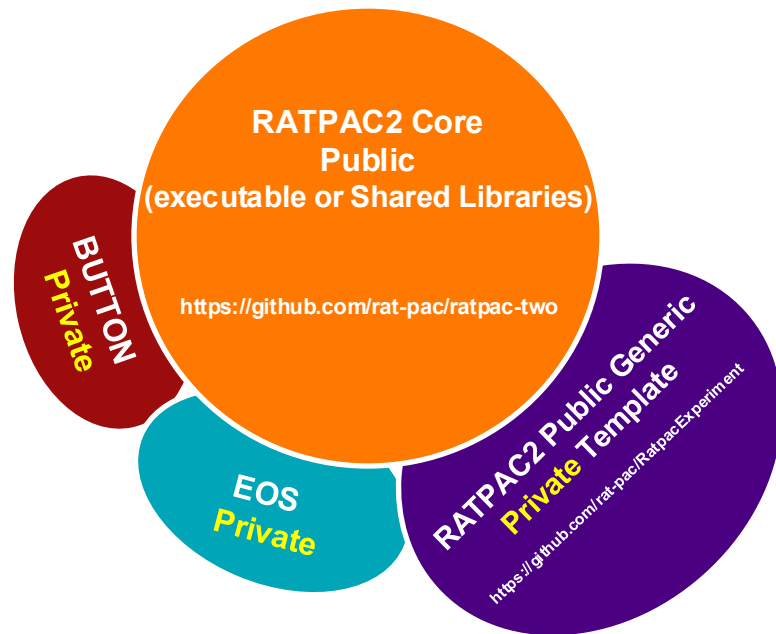


# RATPAC2

## *Current Philosophy*



- To ensure that new features and software can be maintain smoothly, and to prevent forking, the software is now split into two parts
  - RATPAC2 Core : Contains all official software with external dependencies CRY and Tensorflow capabilities, C++17, Geant4 11+ ROOT 6+.
  - RATPAC2 private: Contains any features or software that you would like to keep private (fitter, detector constants, ...)



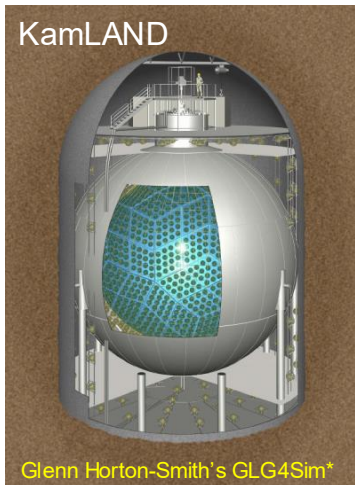
<https://rat-pac.github.io/ratpac2-tutorial/00-setup.html>

# What is inside RATPAC2

*What was shared/used/salvaged from different experiments?*



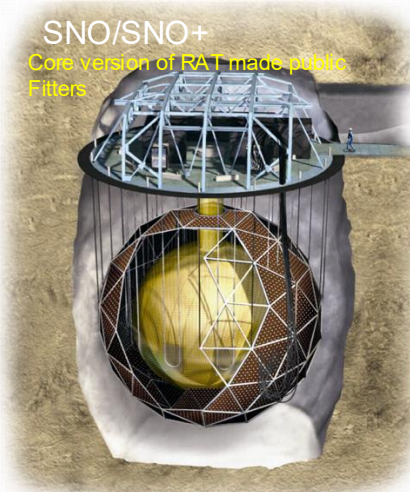
KamLAND



Glenn Horton-Smith's GLG4Sim\*

SNO/SNO+

Core version of RAT made public  
Filters



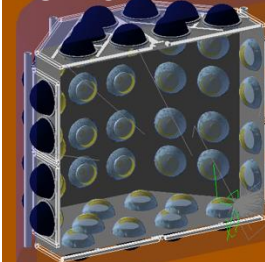
Super Kamiokande



Event reconstruction BONSAI\*\*

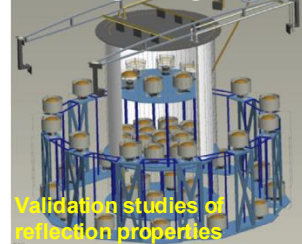
\*\*Currently available as a Pull Request

BUTTON



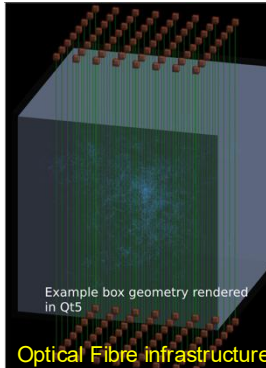
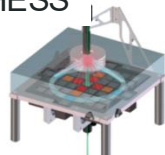
PMT Encapsulation

WATCHBOY



Validation studies of  
reflection properties

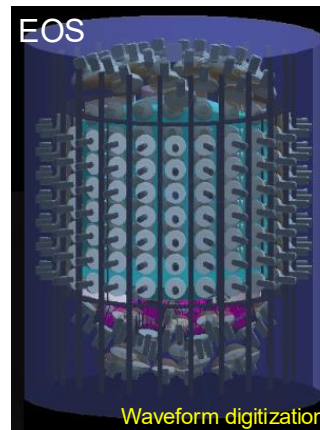
CHESS



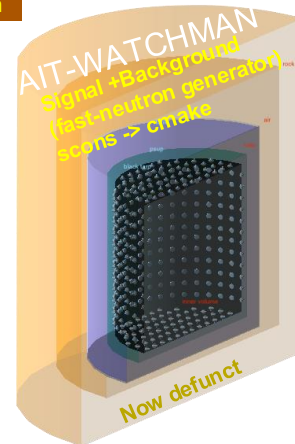
Example box geometry rendered  
in Qt5

Optical Fibre infrastructure

EOS



Waveform digitization



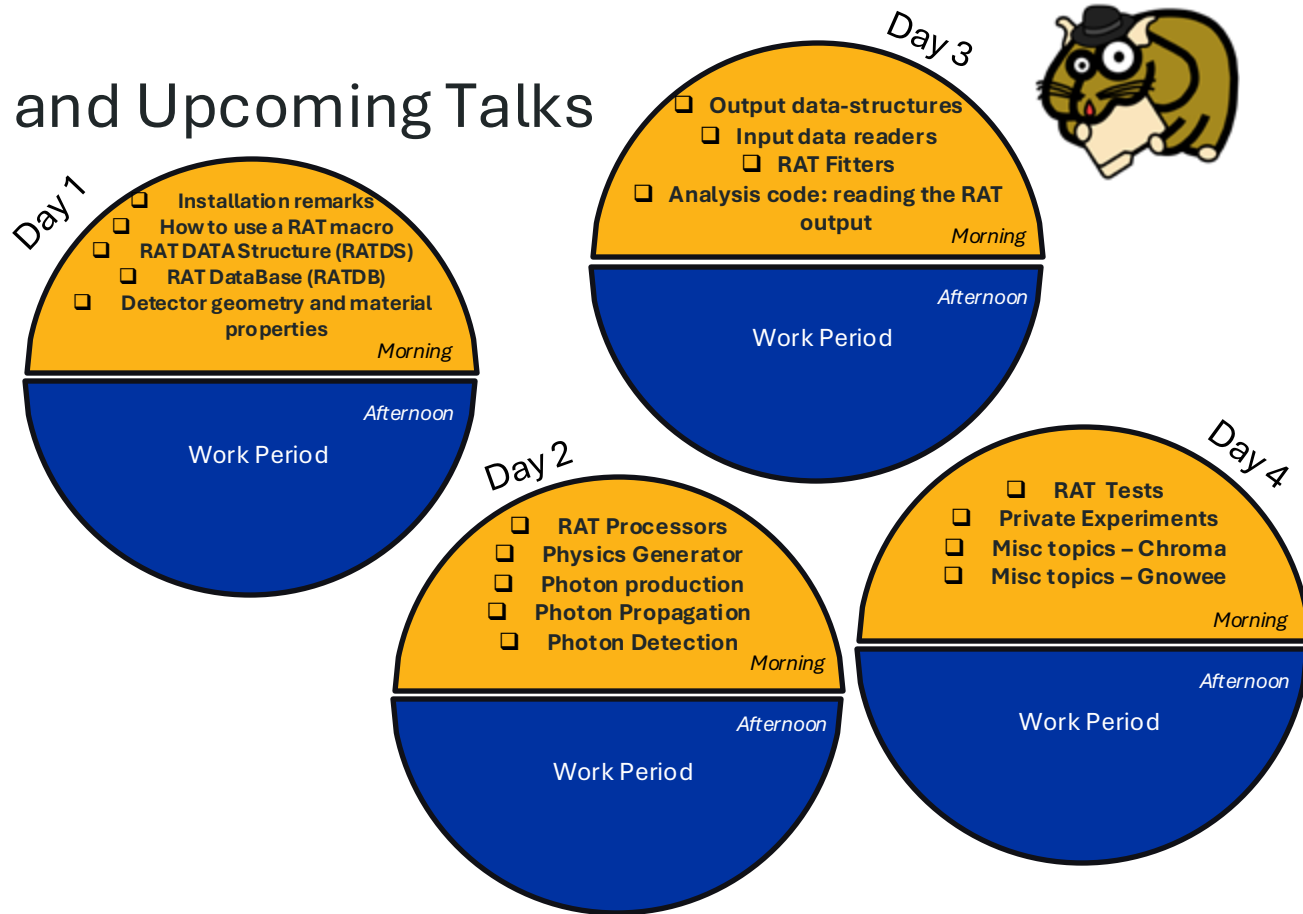
AIT-WATCHMAN  
Signal + Background  
(fast-neutron generator)  
scans -> cmake

Now defunct

\*<https://www.phys.ksu.edu/personal/gahs/GLG4sim/>

# Workshop Format and Upcoming Talks

- Talks in the morning
- Work exercise in afternoon

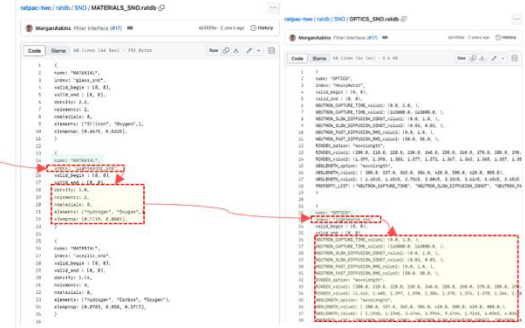
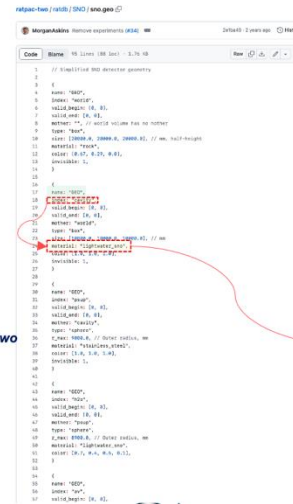
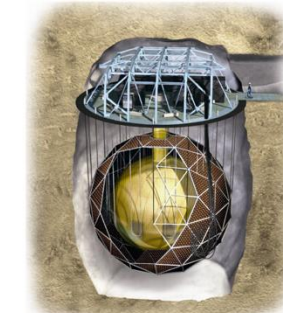


# Day 1: How to define an experiment and its properties



## Morning Talks

0. Introduction and summary – Marc
1. Installation & containers -- James
2. How to use a RAT macro, difference between processor vs. producer (generator) -- Garrett
3. RATDS – Tony
4. RATDB – Marc
5. Detector geometry and material properties (+ geo factories) – James



## HIGHLIGHT

Every component of the simulation is broken down into tables of text entries, in the RATDB folder

We attempt to have no hard coded in values

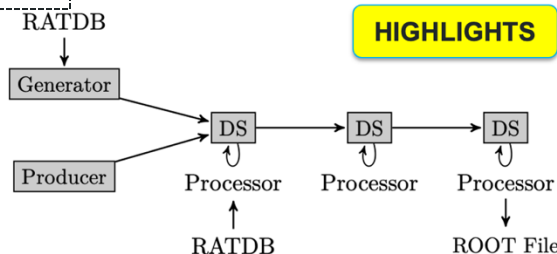
Any of these entries can be changed at the macro stage!



# Day 2: Inner-working of the simulation process

## Morning Talks

1. Physics generators (including Gsim) – Tony
- 2a. Photon production – Tanner
- 2b. Photon (and particle) propagation (+ tracking) – Tanner
3. Photon detection, PE simulation, digitization (including all the PMT pulse stuff) – James
4. RAT processors, including DAQ and analysis, waveform analysis etc. -- Tanner



- **RATDB:** RAT DataBase
- **DS:** Data Structure
- **Generator:** Handles physics generation
- **Producer:** Handles reprocessing of data
- **Processor:** Applies algorithm to data
- **ROOT File:** Output of detector response or actual data

A Few Examples:

`outroot` Write data structure to a ROOT file  
`outnet` Send event to a remote RAT instance  
`simpledaq` A trivial DAQ/trigger simulation where all photoelectrons trigger a PMT hit  
`prune` Trim off unwanted parts of data structure  
`fprompt` Calculate fraction of hits in a prompt time window  
`calibratePMT` Load and apply PMT calibration constants from the database (SNO+)  
`shellFit` GPU-accelerated vertex fitter (MiniCLEAN)

*An event data structure is populated by a generator or producer, and operated on by processors.*

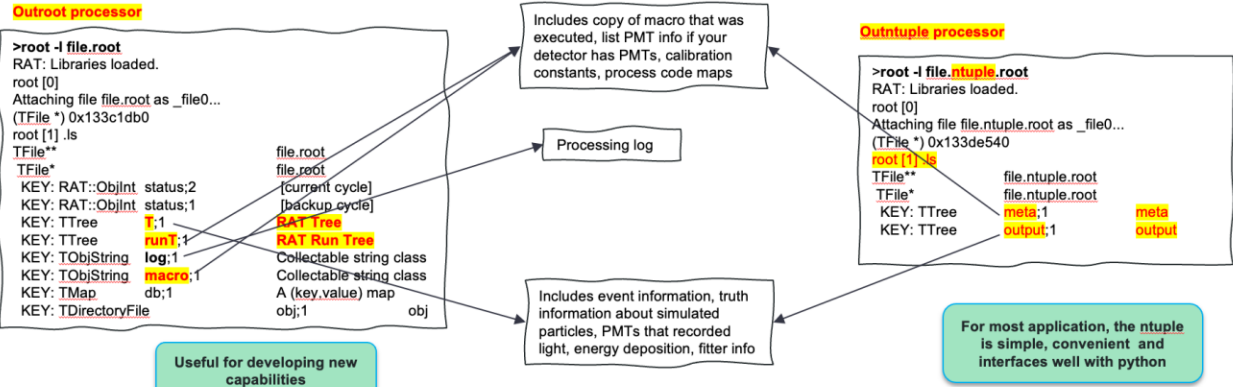
# Day 3: Simulated detector response data, data analysis



## Morning Talks

1. RAT fitters and input handler – Sam
2. Output data-structures (ntuple & root DS & extended ntuple with eos as an example) -- Tony
3. Input data readers (Eos experience with InHDF5) -- James
4. Analysis code: reading the RAT output – Garrett
5. RAT tests – Sam (moved from Day 3)

### HIGHLIGHTS





# Day 4: New features, Maintenances, building powerful new features

## Morning Talks

1. Setting up a private experiment (Eos experience) -- James
2. *RAT tests* – Sam (*moved to Day 3*)
3. Chroma – James
4. Detector design optimization – Emma Houston (presented by Marc)

### Software maintenance

**Question:** How can we ensure that new features do not break anything for other users

**Answer:** rattest & more

### Preventing Split-off

**Question:** How can we ensure that the code does not split-off

**Answer:** ratpac-core and ratpac-private

### Require GPU

**Question:** If you are experiment require fast processing

**Answer:** Chroma

### Detector Optimization

**Question:** If you are experiment require detector design optimization

**Answer:** GNOWEE example





# RATDB

## RAT Data Base

### ratpac-two summer school tutorials

Marc Bergevin

Lawrence Livermore National Laboratory

Prepared by LLNL under Contract DE-AC52-07NA27344.



# Reactor Analysis Tool Data Base (RATDB)

## ● What is a RATDB and why is it useful?

- The files in the **\$RATSHARE/ratdb/** directory are json-style tables that can be easily loaded by the rat source code. The represent all aspect needed to model a detector and its response.
- The files have a very specific syntax that includes a **name**, and **index**, **fields/parameters**, and **values**.
- The files are loaded at run-time and **do not require recompilation of the code**.
- The files contain parameters that you may want to change from the macro. **Any** parameters, including arrays, **can be changed at run time from the macro**:
  - **/rat/db/set NAME[INDEX] FIELD VALUE**
- **NAME**: represents the type of feature we want [material, element, optics, PMTINFO, ...]
- **INDEX**: represent a tag (or index) that other part of the software can reference when they need access to the information located in the **NAME[INDEX]**
- To add a new feature, simply create an entry with a **new index name**. Exercise, add a new material.

*In another file in the RATDB folder*

```
{
  name: "ELEMENT",
  index: "Silicon",
  valid_begin: [0, 0],
  valid_end: [0, 0],
  SYMBOL: "Si",
  z: 14,
  a: 28.09,
}
```

*In a file in the RATDB folder*

```
{
  name: "MATERIAL",
  index: "glass_sno",
  valid_begin: [0, 0],
  valid_end: [0, 0],
  density: 2.2,
  nelements: 2,
  nmaterials: 0,
  elements: ["Silicon", "Oxygen"],
  elemprop: [0.4675, 0.5325],
}
```

*Within a macro [no space in array]*

**/rat/db/set MATERIAL[glass\_sno] elemprop [0.48,0.52]**

or

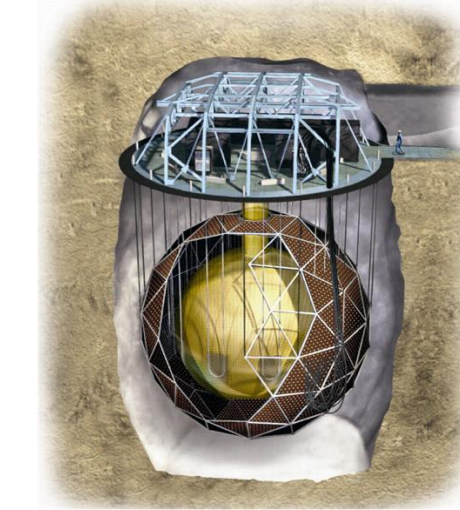
**/rat/db/set ELEMENT[Silicon] a 28.11**

**Note:** We are in the process of removing **valid\_begin** and **valid\_end** for a more intuitive format **PR296**

May also adopt more json format of adding double quotes to field names **"index"** instead of **index**



# Day 1: How to define an experiments and its properties



ratpac-two / ratdb / SNO / <https://github.com/rat-pac/ratpac-two>

| MorganAskins        | Remove experiments (#34) | 2e1ba49 - 2 years ago | History |
|---------------------|--------------------------|-----------------------|---------|
| Name                | Last commit message      | Last commit date      |         |
| ..                  |                          |                       |         |
| MATERIALS_SNO.ratdb | Filter interface (#17)   | 2 years ago           |         |
| OPTICS_SNO.ratdb    | Filter interface (#17)   | 2 years ago           |         |
| PMTINFO.ratdb       | Remove experiments (#34) | 2 years ago           |         |
| sno.geo             | Remove experiments (#34) | 2 years ago           |         |



```

ratpac-two / ratdb / SNO / sno.geo
MorganAskins Remove experiments (#34) 2e1ba49 - 2 years ago History
Code Blame 95 lines (88 loc) · 1.76 KB Raw
1 // Simplified SNO detector geometry
2
3 {
4   name: "GEO",
5   index: "world",
6   valid_begin: [0, 0],
7   valid_end: [0, 0],
8   mother: "", // world volume has no mother
9   type: "box",
10  size: [20000.0, 20000.0, 20000.0], // mm, half-height
11  material: "rock",
12  color: [0.67, 0.29, 0.8],
13  invisible: 1,
14 }
15
16 {
17   name: "GEO",
18   index: "glass_sno",
19   valid_begin: [0, 0],
20   valid_end: [0, 0],
21   mother: "world",
22   type: "box",
23   size: [1000.0, 1000.0, 1000.0], // mm
24   material: "lightwater_sno",
25   color: [0.67, 0.29, 0.8],
26   invisible: 1,
27 }
28
29 {
30   name: "GEO",
31   index: "psup",
32   valid_begin: [0, 0],
33   valid_end: [0, 0],
34   mother: "snavy",
35   type: "sphere",
36   r_max: 9000.0, // Outer radius, mm
37   material: "stainless-steel",
38   color: [1.0, 1.0, 1.0],
39   invisible: 1,
40 }
41
42 {
43   name: "GEO",
44   index: "h2o",
45   valid_begin: [0, 0],
46   valid_end: [0, 0],
47   mother: "psup",
48   type: "sphere",
49   r_max: 8000.0, // Outer radius, mm
50   material: "lightwater_sno",
51   color: [0.7, 0.4, 0.5, 0.1],
52 }
53
54 {
55   name: "GEO",
56   index: "sp",
57   valid_begin: [0, 0],

```

Every component of the simulation is broken down into tables of text entries, in the RATDB folder

We attempt to have no hard coded in values

Any of these entries can be changed at the macro stage!

```

ratpac-two / ratdb / SNO / MATERIALS_SNO.ratdb
MorganAskins Filter interface (#17) ab1666a - 2 years ago History
Code Blame 48 Lines (44 loc) · 792 Bytes Raw
1 {
2   name: "MATERIAL",
3   index: "glass_sno",
4   valid_begin: [0, 0],
5   valid_end: [0, 0],
6   density: 2.2,
7   relements: 2,
8   materials: 0,
9   elements: ["Silicon", "Oxygen"],
10  elempor: [0.4475, 0.5325],
11 }
12
13 {
14   name: "MATERIAL",
15   index: "heavywater",
16   valid_begin: [0, 0],
17   valid_end: [0, 0],
18   density: 1.0,
19   relements: 2,
20   materials: 0,
21   elements: ["Hydrogen", "Oxygen"],
22   elempor: [0.1119, 0.8881],
23 }
24
25 {
26   name: "MATERIAL",
27   index: "heavywater",
28   valid_begin: [0, 0],
29   valid_end: [0, 0],
30   density: 1.14,
31   relements: 0,
32   materials: 0,
33   elements: ["Hydrogen", "Carbon", "Oxygen"],
34   elempor: [0.8783, 0.058, 0.3717],
35 }
36

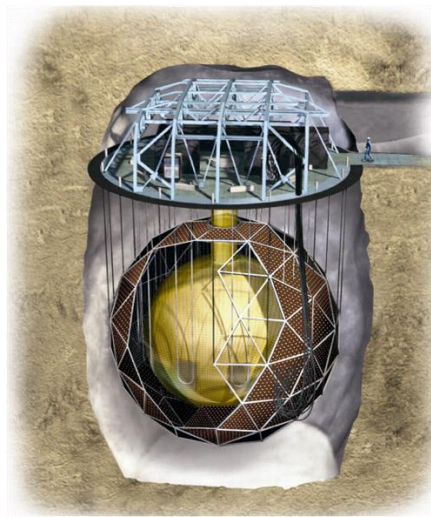
```

```

ratpac-two / ratdb / SNO / OPTICS_SNO.ratdb
MorganAskins Filter interface (#17) ab1666a - 2 years ago History
Code Blame 48 Lines (44 loc) · 5.6 KB Raw
1 {
2   name: "OPTICS",
3   index: "HeavyWater",
4   valid_begin: [0, 0],
5   valid_end: [0, 0],
6   NEUTRON_CAPTURE_TIME_value: [0.0, 1.0, ],
7   NEUTRON_CAPTURE_TIME_value2: [16000.0, 16000.0, ],
8   NEUTRON_SLOW_DIFFUSION_CONST_value: [0.0, 1.0, ],
9   NEUTRON_SLOW_DIFFUSION_CONST_value2: [0.0, 0.0, ],
10  NEUTRON_FAST_DIFFUSION_RMS_value: [0.0, 1.0, ],
11  NEUTRON_FAST_DIFFUSION_RMS_value2: [50.0, 50.0, ],
12  RINDEX_option: "wavelength",
13  RINDEX_value: [200.0, 210.0, 220.0, 230.0, 240.0, 250.0, 260.0, 270.0, 280.0, 290.0],
14  RINDEX_value2: [1.399, 1.399, 1.399, 1.399, 1.399, 1.399, 1.399, 1.399, 1.399, 1.399],
15  ABLENTH_option: "wavelength",
16  ABLENTH_value: [200.0, 337.0, 365.0, 380.0, 420.0, 500.0, 620.0, 800.0],
17  ABLENTH_value2: [1.650, 1.650, 3.745, 3.745, 3.900, 3.945, 3.945, 3.945, 3.945, 3.945],
18  PROPERTY_LIST: ["NEUTRON_CAPTURE_TIME", "NEUTRON_SLOW_DIFFUSION_CONST", "NEUTRON_FAST_DIFFUSION_RMS", "RINDEX", "ABLENTH"],
19 }
20
21 {
22   name: "OPTICS",
23   index: "HeavyWater",
24   valid_begin: [0, 0],
25   valid_end: [0, 0],
26   NEUTRON_CAPTURE_TIME_value: [0.0, 1.0, ],
27   NEUTRON_CAPTURE_TIME_value2: [16000.0, 16000.0, ],
28   NEUTRON_SLOW_DIFFUSION_CONST_value: [0.0, 1.0, ],
29   NEUTRON_SLOW_DIFFUSION_CONST_value2: [0.0, 0.0, ],
30   NEUTRON_FAST_DIFFUSION_RMS_value: [0.0, 1.0, ],
31   NEUTRON_FAST_DIFFUSION_RMS_value2: [50.0, 50.0, ],
32   RINDEX_option: "wavelength",
33   RINDEX_value: [200.0, 210.0, 220.0, 230.0, 240.0, 250.0, 260.0, 270.0, 280.0, 290.0],
34   RINDEX_value2: [1.450, 1.450, 1.450, 1.397, 1.397, 1.397, 1.397, 1.397, 1.397, 1.397],
35   ABLENTH_option: "wavelength",
36   ABLENTH_value: [200.0, 337.0, 365.0, 380.0, 420.0, 500.0, 620.0, 800.0],
37   ABLENTH_value2: [1.150, 1.150, 3.476, 3.476, 3.945, 3.945, 3.945, 3.945, 3.945, 3.945],
38   PROPERTY_LIST: ["NEUTRON_CAPTURE_TIME", "NEUTRON_SLOW_DIFFUSION_CONST", "NEUTRON_FAST_DIFFUSION_RMS", "RINDEX", "ABLENTH"],
39 }

```

# A fuller picture



ratpac-two / ratdb / SNO / <https://github.com/rat-pac/ratpac-two>

MorganAskins Remove experiments (#34) 2e1ba49 · 2 years ago History

| Name                | Last commit message      | Last commit date |
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| ..                  |                          |                  |
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| sno.geo             | Remove experiments (#34) | 2 years ago      |

Experiment  
specific values

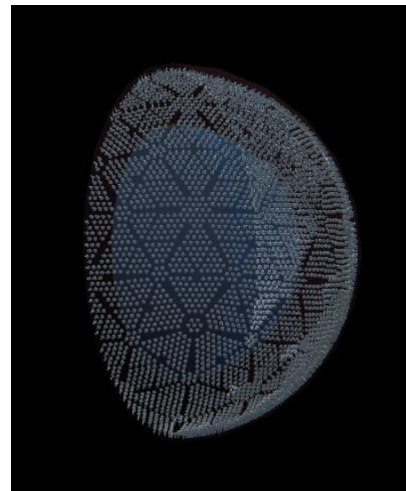
ratpac-two / ratdb /

hbjamin Add stopped pion neutrino source flux as option for generators to sam...

| Name                          |
|-------------------------------|
| ..                            |
| SNO                           |
| SuperK                        |
| Validation                    |
| dichroic_data                 |
| AFTERPULSEProc.ratdb          |
| BNL_WLS_MODEL.ratdb           |
| CHANNEL_STATUS_DEFAULTS.ratdb |
| CRY.ratdb                     |
| DAQ.ratdb                     |
| DETECTOR.ratdb                |
| DICEBOX1586D.ratdb            |
| DIGITIZER.ratdb               |
| DIGITIZER_ANALYSIS.ratdb      |
| Decay0.ratdb                  |
| Decay0Backgr.ratdb            |
| ELEMENTS.ratdb                |
| ESCrossSection.ratdb          |
| FITTER.ratdb                  |
| FIT_QUAD.ratdb                |
| FTP.ratdb                     |
| IBD.ratdb                     |
| IO.ratdb                      |
| LED.ratdb                     |
| MATERIALS.ratdb               |
| MC.ratdb                      |
| NOISEProc.ratdb               |
| OPTICS.ratdb                  |
| OPTICS_Dichroic.ratdb         |
| OPTICS_Photocathode.ratdb     |
| PHOTON_THINNING.ratdb         |
| PHYSICS.ratdb                 |
| PMT.ratdb                     |
| PMTCARGE.ratdb                |
| PMTPULSES.ratdb               |
| PMTTRANSIT.ratdb              |
| QUENCHING.ratdb               |
| RUN.ratdb                     |
| SOLAR.ratdb                   |
| SPECTRUM.ratdb                |
| SPECTRUM_SUPERNOVA.ratdb      |
| STPL.ratdb                    |
| WIMP.ratdb                    |
| WLSExample.ratdb              |
| beta_decays.dat               |
| ratdb_couchdb.json            |

Default values

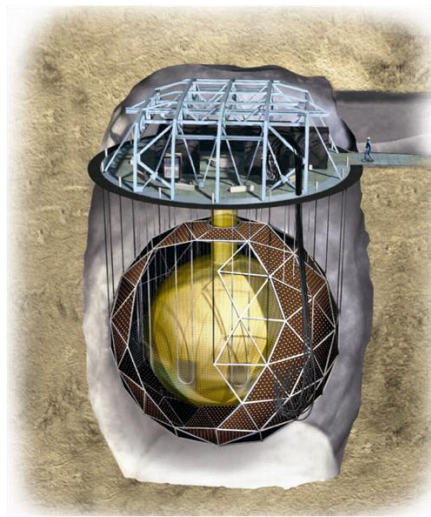
rat --vis \$RATSHARE/macros/visualizations/vis\_sno.mac -o ~/file -l ~/file.l



- The files in the `$RATSHARE/ratdb/` directory are `json`-style tables that can be easily loaded by the rat source code. The represent all aspect needed to model a detector and its response.

Convention is that new experiment be defined as folders in the RATDB folder

# A fuller picture



ratpac-two / ratdb / SNO / <https://github.com/rat-pac/ratpac-two>

MorganAskins Remove experiments (#34) 2e1ba49 · 2 years ago History

| Name                | Last commit message      | Last commit date |
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| sno.geo             | Remove experiments (#34) | 2 years ago      |

Experiment specific values

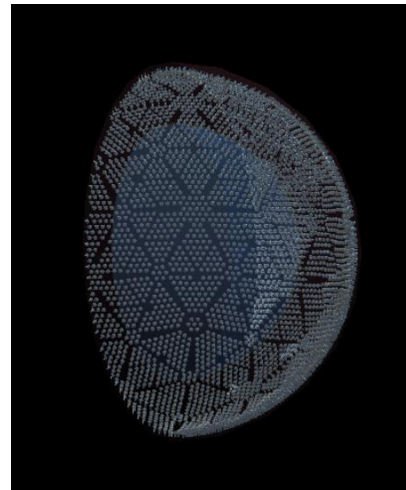
ratpac-two / ratdb /

hbjamin Add stopped pion neutrino source flux as option for generators to sam...

| Name                          |
|-------------------------------|
| ..                            |
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| SuperK                        |
| Validation                    |
| dichroic_data                 |
| AFTERPULSEProc.ratdb          |
| BNL_WLS_MODEL.ratdb           |
| CHANNEL_STATUS_DEFAULTS.ratdb |
| CRY.ratdb                     |
| DAQ.ratdb                     |
| DETECTOR.ratdb                |
| DICEBOX158GD.ratdb            |
| DIGITIZER.ratdb               |
| DIGITIZER_ANALYSIS.ratdb      |
| Decay0.ratdb                  |
| Decay0Backgr.ratdb            |
| ELEMENTS.ratdb                |
| ESCrossSection.ratdb          |
| FITTER.ratdb                  |
| FIT_QUAD.ratdb                |
| FTP.ratdb                     |
| IBD.ratdb                     |
| IO.ratdb                      |
| LED.ratdb                     |
| MATERIALS.ratdb               |
| MC.ratdb                      |
| NOISEProc.ratdb               |
| OPTICS.ratdb                  |
| OPTICS_Dichroic.ratdb         |
| OPTICS_Photocathode.ratdb     |
| PHOTON_THINNING.ratdb         |
| PHYSICS.ratdb                 |
| PMT.ratdb                     |
| PMTCHARGE.ratdb               |
| PMTPULSES.ratdb               |
| PMTTRANSIT.ratdb              |
| QUENCHING.ratdb               |
| RUN.ratdb                     |
| SOLAR.ratdb                   |
| SPECTRUM.ratdb                |
| SPECTRUM_SUPERNOVA.ratdb      |
| STPL.ratdb                    |
| WIMP.ratdb                    |
| WLSExample.ratdb              |
| beta_decay.dat                |
| ratdb_couchdb.json            |

Default values

rat --vis \$RATSHARE/macros/visualizations/vis\_sno.mac -o ~/file -l ~/file.l



PMTs info defined in GEO

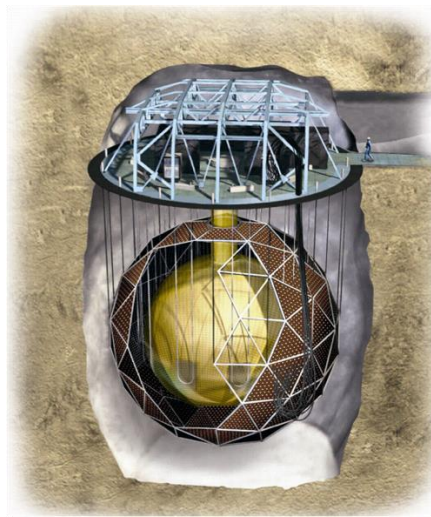
ratpac-two / ratdb / SNO / sno.geo

Code Blame 95 lines (88 loc) · 1.76 KB

```

66 {
67   name: "geo",
68   class: "SuperK",
69   valid_begin: [0, 0],
70   valid_end: [0, 0],
71   mother: "h2o",
72   type: "pmarray",
73   pm_array: {
74     pm_detector_type: "sgm",
75     sensitive_detector: "mydet/pmt/inner",
76     efficiency_correction: 1.027,
77     add_concentrator: 1, // Flag: 0 = no concentrators, 1 = concentrators
78     pos_table: "PMTINFO",
79     orientation: "point", // "manual" means there is a dir_x, dir_y, dir_z
80     // In pos_table for manual orientation of PMTs
81     orient_point: [0.0, 0.0, 0.0],
82   }
83 }
    
```

# A fuller picture



ratpac-two / ratdb / SNO / <https://github.com/rat-pac/ratpac-two>

MorganAskins Remove experiments (#34) 2e1ba49 · 2 years ago History

| Name                | Last commit message      | Last commit date |
|---------------------|--------------------------|------------------|
| ..                  |                          |                  |
| MATERIALS_SNO.ratdb | Filter interface (#17)   | 2 years ago      |
| OPTICS_SNO.ratdb    | Filter interface (#17)   |                  |
| PMTINFO.ratdb       | Remove experiments (#34) |                  |
| sno.geo             | Remove experiments (#34) | 2 years ago      |

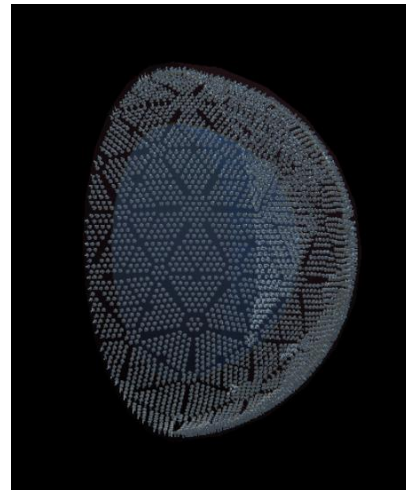
Experiment specific values

ratpac-two / ratdb / [hbjamin](#) Add stopped pion neutrino source flux as option for generators to sam...

| Name                          |
|-------------------------------|
| ..                            |
| SNO                           |
| SuperK                        |
| Validation                    |
| dichroic_data                 |
| AFTERPULSEProc.ratdb          |
| BNL_WLS_MODEL.ratdb           |
| CHANNEL_STATUS_DEFAULTS.ratdb |
| CRY.ratdb                     |
| DAQ.ratdb                     |
| DETECTOR.ratdb                |
| DICEBOX15860.ratdb            |
| DIGITIZER.ratdb               |
| DIGITIZER_ANALYSIS.ratdb      |
| Decay0.ratdb                  |
| Decay0Backgr.ratdb            |
| ELEMENTS.ratdb                |
| ESCrossSection.ratdb          |
| FITTER.ratdb                  |
| FIT_QUAD.ratdb                |
| FTP.ratdb                     |
| IBD.ratdb                     |
| IO.ratdb                      |
| LED.ratdb                     |
| MATERIALS.ratdb               |
| MC.ratdb                      |
| NOISEProc.ratdb               |
| OPTICS.ratdb                  |
| OPTICS_Dichroic.ratdb         |
| OPTICS_Photocathode.ratdb     |
| PHOTON_THINNING.ratdb         |
| PMTINFO.ratdb                 |
| PMT.ratdb                     |
| PMTCHARGE.ratdb               |
| PMTPULSES.ratdb               |
| PMTTRANSIT.ratdb              |
| PMTWAVELENGTH.ratdb           |
| RUN.ratdb                     |
| SOLAR.ratdb                   |
| SPECTRUM.ratdb                |
| SPECTRUM_SUPERNOVA.ratdb      |
| STPL.ratdb                    |
| WIMP.ratdb                    |
| WLSExample.ratdb              |
| beta_decay.dat                |
| ratdb_couchdb.json            |

Default values

rat --vis \$RATSHARE/macros/visualizations/vis\_sno.mac -o ~/file -l ~/file.l



PMTs info defined in GEO

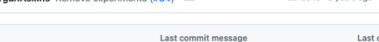
ratpac-two / ratdb / SNO / sno.geo

Code Blame 95 lines (88 loc) · 1.76 KB

```

66 {
67   name: "geo",
68   valid_begin: [0, 0],
69   valid_end: [0, 0],
70   mother: "h2o",
71   type: "pointarray",
72   pmtdetector: "point",
73   sensitive_detector: "mydet/pmt/inner",
74   efficiency_correction: 1.027,
75   add_concentrator: 1, // Flag: 0 = no concentrators, 1 = concentrators
76   // "manual" means there is a dir_x, dir_y, dir_z
77   // in pos_table for manual orientation of PMTs
78   orient_point: [0.0, 0.0, 0.0],
79 }

```



**MorganAskins** Remove experiments (#34) 2e7ba49 · 2 years ago History

| Name                | Last commit message      | Last commit date |
|---------------------|--------------------------|------------------|
| ..                  |                          |                  |
| MATERIALS_SNO.ratdb | Filter interface (#17)   | 2 years ago      |
| OPTICS_SNO.ratdb    | Filter interface (#17)   |                  |
| PMTINFO.ratdb       | Remove experiments (#34) |                  |
| sno.geo             | Remove experiments (#34) | 2 years ago      |

**Experiment specific**

**Experiment  
specific values**

### Default values

```
rat --vis $RATSHARE/macros/visualizations/vis_sno.mac -o ~/file -l ~/file.l
```



Individual PMTs  
position loaded

**PMTs info  
defined in GEO**

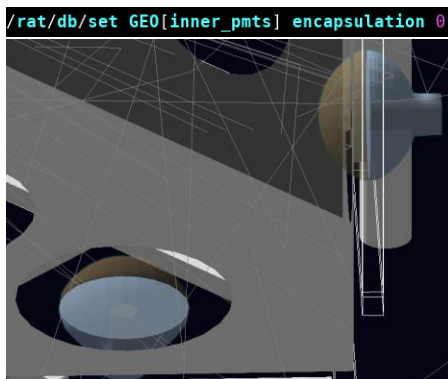
[ratpac-two](#) / [ratdb](#) / [SNO](#) / [sno.geo](#)

**Exercises:** Add a new type of PMTs. Add a new material. Add a new SPECTRUM.

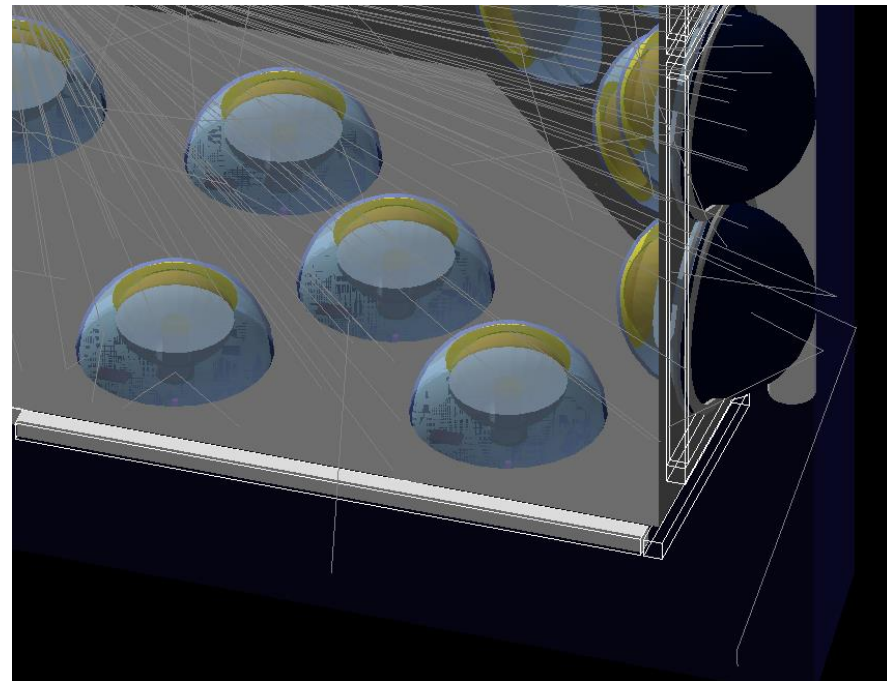
# What if the RATDB feature (name) does not exist?



We present a case study



```
/rat/db/set GEO[inner_pmts] encapsulation 1  
/run/initialize
```



# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

 Open LSexton2396 wants to merge 48 commits

 Conversation 38  Commits 48 

 Changes from all commits  File filter  Conversation

doc/users\_guide

pmt.rst

ratdb

ENCAPSULATION.ratdb

src/geo

CMakeLists.txt

include/RAT

HemisphereEncapsulation.hh

PMTEncapsulation.hh

src/pmt

HemisphereEncapsulation.cc

PMTEncapsulation.cc

PMTFactoryBase.cc

787

788

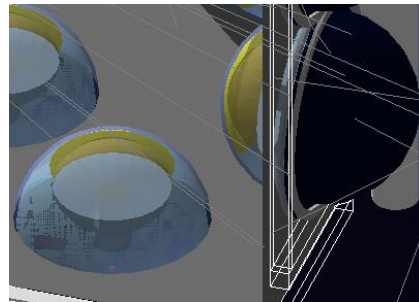
789

790

791

792

- Certain experiments encapsulate their PMT in a dome (BUTTON ICECUBE). This feature was not originally part of ratpac2-core
  - Requires to not interfere with previous code implementation
  - Requires switches to turn feature on or off
  - Require parametrization of encapsulation to enable systematic uncertainty studies and flexibility to be ported to different experiments



# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

[Open](#) LSexton2396 wants to merge 48 commits

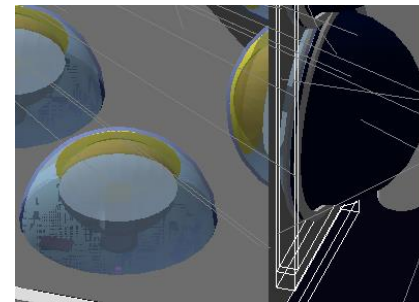
Conversation 38 Commits 48

Changes from all commits File filter Conversation

Filter changed files

- doc/users\_guide
  - pmt.rst
- ratdb
  - ENCAPSULATION.ratdb
- src/geo
  - CMakeLists.txt
- include/RAT
  - HemisphereEncapsulation.hh
  - PMTEncapsulation.hh
- src/pmt
  - HemisphereEncapsulation.cc
  - PMTEncapsulation.cc
  - PMTFactoryBase.cc

Document any changes of any new features



# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

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doc/users\_guide

pmt.rst

ratdb

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include/RAT

HemisphereEncapsulation.hh

PMTEncapsulation.hh

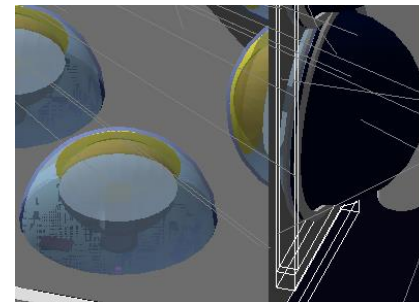
src/pmt

HemisphereEncapsulation.cc

PMTEncapsulation.cc

PMTFactoryBase.cc

This section contains all new parametrization that will become part of RATPAC2-CORE



# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

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pmt.rst

ratdb

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HemisphereEncapsulation.hh

PMTEncapsulation.hh

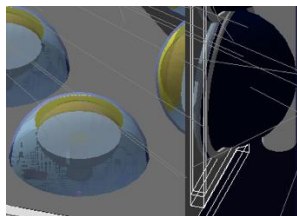
src/pmt

HemisphereEncapsulation.cc

PMTEncapsulation.cc

PMTFactoryBase.cc

This section contains all new parametrization that will become part of RATPAC2-CORE



```
31  ratdb/ENCAPSULATION.ratdb
...  @@ -0,0 +1,31 @@
1  {
2  + name: "ENCAPSULATION",
3  + under: "BUTTON",
4  + valid_begin: [ 0, 0 ],
5  + valid_end: [ 0, 0 ],
6  +
7  + construction: "hemisphere",
8  + envelope_radius: 244.0, // should be large enough to fit all objects (>flange_rmax)
9  + front_encapsulation_material: "nakano_acrylic",
10 + rear_encapsulation_material: "acrylic_black",
11 + flange_rmax: 243.0, // applies to acrylic and metal flange
12 + encap_radius: 200.0,
13 + encap_thickness: 8.0,
14 + use_metal_flange: 1,
15 + metal_flange_material: "stainless_steel",
16 + metal_flange_dimensions: [ 220.0, 243.0, 4.0 ], // G4Tubs dimensions rmin, rmax, z
17 + inside_encapsulation_material: "air",
18 + use_optical_gel: 1,
19 + optical_gel_material: "optical_grease",
20 + optical_gel_sub_height: 150.0, // start from a dome full of gel of size encap_radius, e.g. height of ...
21 + // gel dome left is (encap_radius - optical_gel_sub_height)
22 + pmtposoffset: [ 0.0, 0.0, 98.0 ],
23 + use_silica_bag: 1,
24 + silica_bag_material: "acrylic_white",
25 + silica_bag_dimensions: [ 18.0, 33.0, 3.0 ], // G4Box dimensions x, y, z
26 + silica_bag_position: [ 0.0, 135.0, -74.0 ], // x, y, z
27 + use_cable: 1,
28 + cable_material: "acrylic_black",
29 + cable_dimensions: [ 0.0, 6.5, 45.0 ], // G4Tubs dimensions rmin, rmax, z
30 + cable_position: [ 0.0, 0.0, -152.0 ], // x, y, z
31 + }
```

We require software that will transform these values into geant4 objects that RATPAC can interact with

# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

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HemisphereEncapsulation.hh

PMTEncapsulation.hh

src/pmt

HemisphereEncapsulation.cc

PMTEncapsulation.cc

PMTFactoryBase.cc

This section is where the new c++ code required to implement a new feature is compiled and linked to RATPAC2-CORE.

We require software that will transform these values into geant4 objects that RATPAC can interact with

```
@@ -64,6 +64,8 @@ add_library(geo OBJECT
64 64 src/pmt/RevolutionPMTConstruction.cc
65 65 src/pmt/ToroidalPMTConstruction.cc
66 66 src/pmt/CylindricalPMTConstruction.cc
67 + src/pmt/PMTEncapsulationConstruction.cc
68 + src/pmt/HemisphereEncapsulation.cc
69 69 )
70 70 # Set our include directories
71 71
```

```
@@ -0,0 +1,31 @@
1
2 name: "ENCAPSULATION",
3 under: "BUTTON",
4 + valid_begin : [ 0, 0 ],
5 + valid_end : [ 0, 0 ],
6 +
7 + construction: "hemisphere",
8 + envelope_radius : 244.0, // should be large enough to fit all objects (>flange_rmax)
9 + front_encapsulation_material: "nakano_acrylic",
10 + rear_encapsulation_material: "acrylic_black",
11 + flange_rmax : 243.0, // applies to acrylic and metal flange
12 + encap_radius : 200.0,
13 + encap_thickness : 8.0,
14 + use_metal_flange : 1,
15 + metal_flange_material : "stainless_steel",
16 + metal_flange_dimensions : [ 220.0, 243.0, 4.0 ], // G4Tubs dimensions rmin, rmax, z
17 + inside_encapsulation_material : "air",
18 + use_optical_gel : 1,
19 + optical_gel_material: "optical_grease",
20 + optical_gel_sub_height : 150.0, // start from a dome full of gel of size encap_radius, e.g. height of ...
21 + // gel dome left is (encap_radius - optical_gel_sub_height)
22 + pmtposoffset : [ 0.0, 0.0, 98.0 ],
23 + use_silica_bag : 1,
24 + silica_bag_material : "acrylic_white",
25 + silica_bag_dimensions : [ 18.0, 33.0, 3.0 ], // G4Box dimensions x, y, z
26 + silica_bag_position: [ 0.0, 135.0, -74.0 ], // x, y, z
27 + use_cable : 1,
28 + cable_material : "acrylic_black",
29 + cable_dimensions : [ 0.0, 6.5, 45.0 ], // G4Tubs dimensions rmin, rmax, z
30 + cable_position : [ 0.0, 0.0, -152.0 ], // x, y, z
31 +
```

# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

[Open](#) LSexton2396 wants to merge 48 commits

Conversation 38 Commits 48

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doc/users\_guide

pmt.rst

ratdb

ENCAPSULATION.ratdb

src/geo

CMakeLists.txt

include/RAT

HemisphereEncapsulation.hh

PMTEncapsulation.hh

src/pmt

HemisphereEncapsulation.cc

PMTEncapsulation.cc

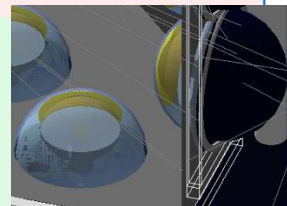
PMTFactoryBase.cc

Modify existing file to add features

```
src/geo/CMakeLists.txt
@@ -64,6 +64,8 @@ add_library(geo OBJECT
64 64 src/pmt/RevolutionPMTConstruction.cc
65 65 src/pmt/ToroidalPMTConstruction.cc
66 66 src/pmt/CylindricalPMTConstruction.cc
67 67 + src/pmt/PMTEncapsulationConstruction.cc
68 68 + src/pmt/HemisphereEncapsulation.cc
69 69 )
70 70 # Set our include directories
```

This section is where the actual new c++ code that links the new GEANT4 object to the PMT

```
src/geo/src/pmt/PMTFactoryBase.cc
@@ -8,6 +8,7 @@
8 8 #include <RAT/Log.hh>
9 9 #include <RAT/Materials.hh>
10 10 #include <RAT/PMTConstruction.hh>
11 11 + #include <RAT/PMTEncapsulationConstruction.hh>
12 12 #include <RAT/PMTFactoryBase.hh>
13 13 #include <RAT/PMTInfoParser.hh>
14 14 #include <algorithm>
@@ -45,8 +46,55 @@ G4VPhysicalVolume *PMTFactoryBase::ConstructPMTs(
45 46 }
46 47 G4ThreeVector local_offset = PMTInfoParser::ComputeLocalOffset(mother_name);
47 48
48 48 - PMTConstruction *construction = PMTConstruction::NewConstruction(lpmt, mother);
49 49 - G4LogicalVolume *log_pmt = construction->BuildVolume(volume_name);
49 49 + PMTEncapsulationConstruction *encap_construction = 0;
50 50 + G4LogicalVolume *log_encapenv = 0;
51 51 + PMTConstruction *construction = 0;
52 52 + G4LogicalVolume *log_pmt = 0;
53 53 + G4VPhysicalVolume *phys_mother_encap = 0;
54 54 + G4ThreeVector pmtoffsetposition(0.0, 0.0, 0.0);
55 55 +
56 56 + int encapsulation = 0; // default to no encapsulation
57 57 +
58 58 + encapsulation = table->GetI("encapsulation");
59 59 + if (encapsulation == 1) {
60 60 +
61 61 + if (encapsulation == 1) {
62 62 + std::string encapsulation_model = table->GetS("encapsulation_model"); // get encapsulation model e.g. BUTTON
63 63 + DBLinkPtr lencapsulation = DB::GetLink(encapsulation_model); // find model in ENCAPSULATION.RATDB, need to make this file
64 64 + "ENCAPSULATION", encapsulation_model);
65 65 +
66 66 + info << "Encapsulation is turned on, using model: " << encapsulation_model << newline;
67 67 +
```



# What if the RATDB feature does not exist?

Case Study, PR #256 (Lewis Sexton, Sheffield University for BUTTON Collaboration)



## PMT encapsulation #256

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PMTEncapsulation.hh

src/pmt

HemisphereEncapsulation.cc

PMTEncapsulation.cc

PMTFactoryBase.cc

This section is where the actual new c++ code that links the new GEANT4 object to the PMT

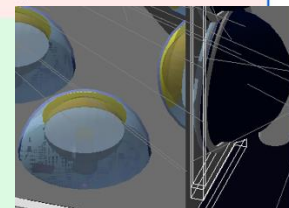
**TIP:** the **GetI** (Get Integer), **GetD** (Get Double), **GetS** (Get String) are the tools used to pull the RATDB values from the tables

Modify existing file to add features

```
src/geo/CMakeLists.txt
@@ -64,6 +64,8 @@ add_library(geo OBJECT
64 64 src/pmt/RevolutionPMTConstruction.cc
65 65 src/pmt/ToroidalPMTConstruction.cc
66 66 src/pmt/CylindricalPMTConstruction.cc
67 67 src/pmt/PMTEncapsulationConstruction.cc
68 68 src/pmt/HemisphereEncapsulation.cc
67 69 )
68 70
69 71 # Set our include directories
```

```
src/geo/src/pmt/PMTFactoryBase.cc
@@ -8,6 +8,7 @@
8 8 #include <RAT/Log.hh>
9 9 #include <RAT/Materials.hh>
10 10 #include <RAT/PMTConstruction.hh>
11 11 #include <RAT/PMTEncapsulationConstruction.hh>
12 12 #include <RAT/PMTFactoryBase.hh>
13 13 #include <RAT/PMTInfoParser.hh>
14 14 #include <algorithm>

@@ -45,8 +46,55 @@ G4VPhysicalVolume *PMTFactoryBase::ConstructPMTs(
45 46 }
46 47 G4ThreeVector local_offset = PMTInfoParser::ComputeLocalOffset(mother_name);
47 48
48 48 - PMTConstruction *construction = PMTConstruction::NewConstruction(lpmt, mother);
49 49 - G4LogicalVolume *log_pmt = construction->BuildVolume(volume_name);
49 49 + PMTEncapsulationConstruction *encap_construction = 0;
50 50 + G4LogicalVolume *log_encapenv = 0;
51 51 + PMTConstruction *construction = 0;
52 52 + G4LogicalVolume *log_pmt = 0;
53 53 + G4VPhysicalVolume *phys_mother_encap = 0;
54 54 + G4ThreeVector pmtoffsetposition(0.0, 0.0, 0.0);
55 55 +
56 56 + int encapsulation = 0; // default to no encapsulation
57 57 + encapsulation = table->GetI("encapsulation");
58 58 + if (encapsulation == 1) {
59 59 + std::string encapsulation_model = table->GetS("encapsulation_model"); // get encapsulation model e.g. BUTTON
60 60 + DBLinkPtr lencapsulation = DB::GetLink("ENCAPSULATION", encapsulation_model); // find model in ENCAPSULATION.RATDB, need to make this file
61 61 + info << "Encapsulation is turned on, using model: " << encapsulation_model << newline;
62 62 +
63 63 +
64 64 +
65 65 +
66 66 +
67 67 +
```



# Summary



**RATDB handles  
parametrization for  
RATPAC2**

**How blocks of text are read  
depend on the defined  
name**

**If a new capability is not  
already coded in is needed,  
we must add it.**

**We only need to do this  
once**

name: "MATERIAL",

name: "GEO",

name: "OPTICS",

name: "PMTINFO",

name: "ENCAPSULATION",

